

SIMRAN KHANNA

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Technical Skills

Languages and Databases: Java, Python, HTML, CSS, JavaScript, MySQL, Oracle

Technologies: Android Development, Cloud Computing, Artificial Intelligence, Web Development

Tools: OpenCV, Software Engineering, Performance Optimization, Selenium

Other Skills: Teamwork, Project Coordination, Communication, Root Cause Analysis

Training/Projects

Hostel Management Application | Developer

May 2024 – June 2024

- Designed and developed an Android application using **Java**, **XML**, and **MySQL**, managing over 1,000 hostel user records securely.
- Integrated RESTful APIs and a **secure Sign-Up feature**, improving real-time **data retrieval by 25%** and ensuring safe user authentication.
- Enhanced database efficiency by **optimizing SQL queries**, achieving a 30% reduction in query execution time and improved data consistency.
- Performed extensive testing with 50+ cases, resolving critical bugs and **increasing application reliability by 20%**.

Automate Media Volume Using Hand Gesture | Developer

January 2024 – February 2024

- Built a **Python-based** system using **OpenCV** to automate PC media volume control via hand gestures, achieving **95% gesture detection accuracy** and distinguishing between 10+ unique gestures.
- Reduced response time by 20%** through algorithm enhancements and optimized image processing pipelines, lowering CPU usage by 15%.
- Tested and debugged the system on 5+ devices, ensuring **robust performance** and cross-platform compatibility.
- Enhanced system efficiency by **optimizing image processing pipelines**, reducing usage of CPU by 15%.

Android Application Development Training

August 2023 – September 2023

Trainee

Chandigarh, India

- Completed a **21-day** intensive training program focused on **Android application development** using **Java**, **XML**, and **MySQL**.
- Built **functional Android application** with **user authentication** and **database integration** handling over 500 user records enhancing data security and real-time data management.
- Applied **RESTful APIs** for seamless data retrieval reducing data fetch time by 30% and improved **user experience** through optimized interface design.
- Gained proficiency in **Android Studio**, **UI/UX design principles**, **debugging techniques**, and **performance optimization** reducing application crashes by 25% during testing.

Certifications & Accomplishments

- Core Java Certification**, Hacker Rank, 2024 (Top 10% of learners) and earned **5-star badges on Java**, **4-star badges on Issue Resolution**, showcasing strong programming proficiency and Root Cause Analysis capabilities.
- AWS Certified Data Analytics and Data Engineering**, also **Cloud Foundations 2023** – Demonstrated expertise in handling large-scale data pipelines, implementing data analytics solutions, and leveraging AWS cloud infrastructure for efficient data processing.
- Secured **1st position** in the **Code-o-Fiesta** coding event, 2023, organized technical events for **National Science Day 2023**, actively participated in technical competitions at **Parivartan Fest**.

Education

Chandigarh Group of Colleges, Landran

August 2021 – May 2025

Bachelor of Technology (B-Tech) in Computer Science Engineering

Mohali, Punjab

- CGPA:** 8.6/10
- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Database Management Systems, Operating Systems